

# Enhanced Attack Report

## Jaguar Tooth

*Generated on 2025-10-08*

## **Disclaimer**

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## Definitions

**Pre-Conditions:** Conditions that must be true to execute the attack steps in the milestone.

**Post-Conditions:** Traces that an attacker leaves behind after executing the attack steps in the milestone.

**Attack Steps:** Steps that an attacker would take to achieve the goal of the milestone.

**MITRE Technique:** Techniques from the MITRE ATT&CK; framework that are relevant to the milestone.

## STIX

**Malware Name:** Jaguar Tooth

**Malware Description:** Jaguar Tooth is non-persistent malware that targets Cisco IOS routers. Collects device information and exfiltrates over Trivial File Transfer Protocol (TFTP). Enables unauthenticated backdoor access. It is deployed and executed via exploitation of the patched Simple Network Management Protocol (SNMP) vulnerability CVE-2017-6742.

## Quick Overview

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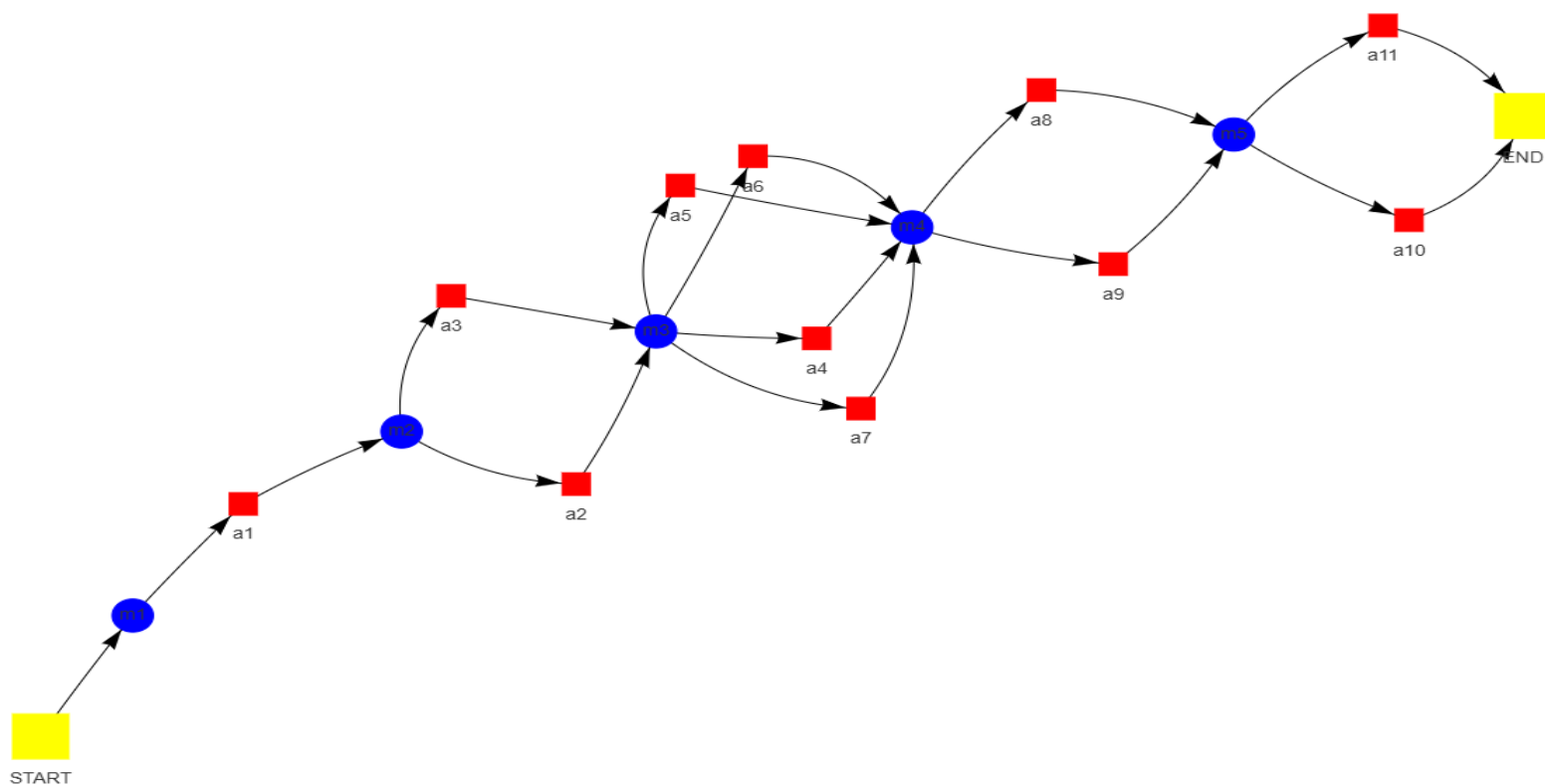
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## Attack Graph



## Milestone *m1*

### Attack Step *a1*

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**Name:** Exploit Public-Facing Application as used by the malware

**Description:** The "Jaguar Tooth" malware exploits vulnerabilities inherent in Cisco router implementations of the Simple Network Management Protocol (SNMP). A known vulnerability within Cisco IOS and IOS XE software (CVE-2017-6742) is leveraged by the malware to facilitate initial access to vulnerable routers. Subsequent exploitation of an authentication-related vulnerability likely enables privilege escalation, granting the malware full administrative control over the compromised device. Information pertaining to the router and its connected network is gathered through the execution of command-line interface (CLI) commands. This data encompasses MAC addresses of networked devices, router configuration details, and software version information. The exfiltration of this sensitive information is likely accomplished via trivial file transfer protocol (TFTP) or covert channels. A persistent backdoor is established on the compromised router, enabling attackers to maintain remote access even after the initial exploit is remediated. This backdoor facilitates further malicious activities such as launching additional attacks from the compromised router, exfiltrating additional data, and utilizing the router as a platform for attacking other devices within the network. Mitigation strategies recommended by the National Cyber Security Centre (NCSC) and Cybersecurity and Infrastructure Security Agency (CISA) include: The immediate application of the latest security updates from Cisco to address vulnerabilities exploited by "Jaguar Tooth" (CVE-2017-6742). Implementation of secure SNMP configuration practices, including disabling unnecessary SNMP services, utilizing strong, unique

passwords for SNMP community strings, configuring access control lists (ACLs) to restrict SNMP access to authorized users and devices, and enabling SNMPv3 with authentication and encryption. Disabling legacy protocols such as Telnet in favor of SSH for secure remote management.

## **Pre-Conditions:**

- The malware has code capable of exploiting the vulnerability.
- A vulnerable public-facing application is accessible.

## **Post-Conditions:**

- Backdoor executables or scripts on the affected routers.
- Data exfiltration from compromised routers.
- Log entries indicating unauthorized access attempts and successful logins.
- Installation of persistent backdoors for future access.
- New user accounts with elevated privileges.
- Modified router configuration files.
- Firewall logs showing bypassed security rules or unusual port activity.
- Altered system timestamps or evidence of time manipulation.
- Complete compromise of affected Cisco routers.
- Denial-of-service attacks against internal or external networks.
- Manipulation of router configurations and traffic routing.
- Unusual network traffic patterns, including data transfers to unknown IP addresses.

## ***MITRE Technique***

**ID:** T1190

**Name:** Exploit public-facing application

**Description:** Adversaries may attempt to exploit a weakness in an Internet-facing host or system to initially access a network. The weakness in the system can be a software bug, a temporary glitch, or a misconfiguration.

**More info:** <https://attack.mitre.org/techniques/T1190/>

# Milestone *m2*

## Attack Step a2

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**Name:** Modify Authentication Process as used by the malware

**Description:** Cisco IOS device authentication processes are subject to modification by the sophisticated malware known as Jaguar Tooth. This malware exploits vulnerabilities within the Cisco IOS operating system to achieve persistent access. Specifically, Jaguar Tooth targets Telnet Authentication and Physical Session Authentication functions. Malicious code is injected into these functions, effectively bypassing standard security checks. Password verification mechanisms are circumvented following code patching, enabling unauthorized access via Telnet or physical console connections without valid credentials. Local accounts on the compromised device are granted elevated privileges, potentially allowing for complete control of the router or switch. The consequences of this modification include unrestricted remote access via Telnet, exposing sensitive information and configuration settings. Physical device compromise is also facilitated, enabling unauthorized control through direct physical access. Furthermore, the compromised device can be leveraged as a platform for lateral movement within the network.

### Pre-Conditions:

- Knowledge of the malware's code and its authentication mechanism.
- Tools capable of modifying the malware's code (e.g., a debugger, disassembler).
- Understanding of programming languages used in the malware.
- Access to the infected system where the malware is running.

### Post-Conditions:

- New user accounts or permissions added
- Unlawful remote access to compromised router
- Potential for further malware deployment and lateral movement within the network
- Firewall rules changes allowing outbound connections to suspicious IPs
- Network traffic logs showing data transfer to external IP addresses
- Modified router configuration files
- Data exfiltration from compromised router
- Logs indicating unauthorized access attempts and successful logins
- Altered system timestamps
- Compromised Cisco router functionality
- Unusual activity in system event logs

### *MITRE Technique*

**ID:** T1556.004

**Name:** Modify authentication process: network device authentication

**Description:** Adversaries may use Patch System Image to hard code a password in the operating system, thus bypassing of native authentication mechanisms for local accounts on network devices.

**More info:** <https://attack.mitre.org/techniques/T1556/004>

## Attack Step a3

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**Name:** Modify System Image: Patch System Image as used by the malware

**Description:** The "Jaguar Tooth" malware exploits a vulnerability within the SNMP protocol processing of susceptible Cisco IOS software versions (specifically, those vulnerable to CVE-2017-6742). Upon successful exploitation, malicious code is injected directly into the running memory of the Cisco IOS system image. This injected code targets authentication routines employed by the router's login mechanisms, resulting in modifications that circumvent standard password checks and user verification processes. The compromised router can then be utilized for unauthorized access, enabling activities such as data exfiltration, establishment of command and control (C2) backdoors, and launching denial-of-service (DDoS) attacks against external targets. Mitigation strategies include the immediate application of security updates from Cisco to address vulnerabilities like CVE-2017-6742, implementation of robust SNMP configurations with strong passwords, access limitations, and encryption (SNMPv3), and network segmentation practices to isolate critical infrastructure.

### Pre-Conditions:

- The system image used by the malware is accessible.
- Access to the target system running the vulnerable software.
- Administrative privileges on the target system.
- Knowledge of the specific vulnerability and the patch required to address it.
- A patched system image is available.

### Post-Conditions:

#### **MITRE Technique**

**ID:** T1601.001

**Name:** Modify system image: patch system image

**Description:** Adversaries may modify the operating system of a network device to introduce new capabilities or weaken existing defenses. Some network devices are built with a monolithic architecture, where the entire operating system and most of the functionality of the device is contained within a single file. Adversaries may change this file in storage, to be loaded in a future boot, or in memory during runtime.

**More info:** <https://attack.mitre.org/techniques/T1601/001>

# Milestone *m3*

## Attack Step *a4*

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**Name:** Remote System Discovery as used by the malware

**Description:** APT28 (also known as Fancy Bear or Cozy Bear) is a highly sophisticated cyber espionage group of Russian origin. It is widely recognized for its persistent targeting of government agencies, political organizations, and critical infrastructure globally. APT28's operations are characterized by advanced techniques and the ability to maintain undetected presence over extended periods. "Jaguar Tooth" is a custom malware specifically developed by APT28 to exploit vulnerabilities inherent in Cisco IOS devices (routers). This malware poses a significant threat due to its capabilities: exploitation of CVE-2017-6742, enabling arbitrary code execution on vulnerable routers; memory injection, facilitating the concealment of malicious payloads within router memory; and establishment of persistent backdoors, granting APT28 unauthorized access and control over compromised devices. The functionalities of "Jaguar Tooth" encompass: exfiltration of sensitive information pertaining to targeted routers through Cisco IOS CLI commands (e.g., show running-config, show version), subsequently transmitted via TFTP; and modification of the router's authentication process, enabling unauthenticated access to local accounts via Telnet and physical connections. The implications of "Jaguar Tooth" are far-reaching: espionage activities such as eavesdropping on network traffic, data theft, and reconnaissance; disruption of network services or attacks against interconnected systems; and utilization of compromised routers as stepping stones for further attacks or as Command and Control (C2) infrastructure for other malware. Mitigation strategies to counter the threat posed by "Jaguar Tooth" include: immediate application of security updates from Cisco to address CVE-2017-6742 and other known vulnerabilities; implementation of strong passwords and multi-factor authentication for all router accounts; network segmentation to isolate critical devices and networks, thereby limiting the impact of a breach; deployment of Intrusion Detection and Prevention Systems (IDPS) to detect and block suspicious activity on the network; and regular security audits to identify vulnerabilities and ensure the effectiveness of security controls.

### Pre-Conditions:

- Network connectivity between the attacker's machine and the target Cisco router.

### Post-Conditions:

- Modified system files containing malware remnants
- Network disruption and service outages
- Presence of malware executables in router memory
- System event logs recording unusual activity
- Firewall logs showing suspicious connections
- Data exfiltration from compromised routers
- Modified Cisco IOS image files
- Increased risk of further attacks
- Loss of sensitive information
- New configuration files with malicious code
- Network traffic analysis showing data transfer to external servers
- Altered SNMP logs indicating unauthorized access

## **MITRE Technique**

**ID:** T1018

**Name:** Remote system discovery

**Description:** Adversaries may attempt to get a listing of other systems by IP address, hostname, or other logical identifier on a network that may be used for Lateral Movement from the current system. Functionality could exist within remote access tools to enable this, but utilities available on the operating system could also be used such as Ping, net view using Net, or, on ESXi servers, esxcli network diag ping.

**More info:** <https://attack.mitre.org/techniques/T1018/>

## **Attack Step a5**

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**Name:** File and Directory Discovery as used by the malware

**Description:** File and directory discovery is performed by the "Jaguar Tooth" malware through the execution of Cisco IOS CLI commands. These commands, such as "dir" or "show running-config," enumerate the contents of the local flash filesystem. The output generated by these commands is captured and analyzed by the malware to identify potential targets. Analysis of the collected data may include the identification of configuration files containing sensitive information, executable files susceptible to modification or exploitation, and other files relevant to the malware's objectives. This action facilitates a comprehensive understanding of the file system structure and contents on the infected router, enabling subsequent malicious activities such as data exfiltration, privilege escalation, or persistence.

### **Pre-Conditions:**

- A compromised Cisco router with the "Jaguar Tooth" malware installed.
- The target system has files and directories present.
- The malware must have access to the file system of the compromised router.
- The malware is running on a target system.
- The target system is connected to a network.

### **Post-Conditions:**

## **MITRE Technique**

**ID:** T1083

**Name:** File and directory discovery

**Description:** Adversaries may enumerate files and directories or may search in specific locations of a host or network share for certain information within a file system. Adversaries may use the information from File and Directory Discovery during automated discovery to shape follow-on behaviors, including whether or not the adversary fully infects the target and/or attempts specific actions.

**More info:** <https://attack.mitre.org/techniques/T1083/>

## **Attack Step a6**

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**Name:** System Network Configuration Discovery as used by the malware

**Description:** The "System Network Configuration Discovery" action undertaken by the Jaguar Tooth malware involves the exploitation of Cisco IOS CLI commands to acquire sensitive system network configuration data. Specific commands employed include, but are not limited to, "show running-config," "show interfaces," "show ip route," and "show vlan." The extracted data is subsequently processed by the malware, with potential outcomes including local storage on the infected router, transmission to a command-and-control server, or utilization for further malicious activities such as network mapping and vulnerability identification. This action poses significant risks to compromised networks. Attackers can leverage gathered configuration information to construct detailed network topologies, pinpoint critical systems, and plan subsequent attacks. Furthermore, the identification of open ports and services facilitates targeted exploitation of vulnerabilities. The potential for credential theft from configuration files containing sensitive usernames and passwords must also be considered, enabling lateral movement within the compromised network.

## Pre-Conditions:

- The attacker possesses knowledge of the vulnerable SNMP OIDs exploited by the malware.
- The target system has SNMP enabled.
- The target system is running a vulnerable version of Cisco IOS or IOS XE software.
- An attacker can establish a network connection to the target system.

## Post-Conditions:

- System instability
- Modified system configuration files
- Malware persistence
- Reputational damage
- Altered firewall rules
- Presence of malware executables or scripts
- Unusual log entries (e.g., failed login attempts, unauthorized access)
- Backdoor creation in system binaries
- Network traffic to malicious servers
- Data exfiltration
- Data files transferred to external locations
- New user accounts created
- Network disruption
- Compromised system control

## MITRE Technique

**ID:** T1016.001

**Name:** System network configuration discovery: internet connection discovery

**Description:** Adversaries may check for Internet connectivity on compromised systems. This may be performed during automated discovery and can be accomplished in numerous ways such as using Ping, tracer, and GET requests to websites.

**More info:** <https://attack.mitre.org/techniques/T1016/001>

## Attack Step a7

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**Name:** System Information Discovery as used by the malware

**Description:** System Information Discovery by the "Jaguar Tooth" Malware on Compromised Cisco Devices: A Technical Analysis The "Jaguar Tooth" malware exploits vulnerabilities within compromised Cisco devices to acquire sensitive system information. A key technique employed is the exploitation of CVE-2017-6742, a buffer overflow vulnerability that permits arbitrary code execution within the device's memory. Subsequently, the malware injects custom shellcode into the system. This injected shellcode functions to incrementally write its payload into designated memory locations, thereby obscuring it from conventional detection methods. While the precise commands utilized by "Jaguar Tooth" for system information gathering remain unspecified, common reconnaissance practices suggest potential targets:

- \* The "show version" command is likely employed to retrieve fundamental details regarding the Cisco IOS software version, hardware platform, and configuration settings.
- \* The "show interfaces" command may be used to obtain a comprehensive inventory of network interfaces (Ethernet ports, serial lines), their operational status (up/down), IP addresses, MAC addresses, and bandwidth utilization.
- \* The "show running-config" command is potentially leveraged to display the current running configuration of the device, revealing settings pertinent to routing protocols, access lists, VLANs, and other critical configurations. The acquisition of this system information serves several purposes for malicious actors:
- \* Vulnerability assessment can be facilitated by identifying the software version, enabling attackers to determine the presence of exploitable vulnerabilities on the specific device.
- \* Network mapping can be achieved through interface information, allowing attackers to delineate the network topology and identify connected devices and potential attack vectors.
- \* Configuration exploitation becomes possible by accessing the running configuration, revealing sensitive settings that can be manipulated for further control or data exfiltration.

Mitigation strategies against such attacks include:

- \* Prompt application of Cisco security updates to address known vulnerabilities such as CVE-2017-6742.
- \* Implementation of robust access control mechanisms, restricting access to management interfaces (SSH, SNMP) through strong authentication and authorization protocols.
- \* Network segmentation practices that isolate critical devices from less secure networks to minimize the impact of a potential breach.
- \* Deployment of intrusion detection/prevention systems (IDS/IPS) capable of detecting suspicious network activity and blocking malicious traffic.

## Pre-Conditions:

- An attacker has knowledge of the vulnerability (CVE-2017-6742).
- The target system is running Cisco IOS or Cisco IOS XE Software.
- Pre-Conditions:
- Knowledge of SNMP protocols and community strings.
- An attacker with network access to the target system.
- The target system has an active SNMP service enabled.
- Requirements:

## Post-Conditions:

- Firewall rules modifications
- Compromised system integrity
- Presence of malicious executables or scripts
- Modified Cisco IOS configuration files
- Unusual network traffic logs (e.g., connections to suspicious IP addresses)
- Altered system event logs
- Loss of sensitive information
- Data exfiltration
- System instability and potential crashes
- Remote code execution capabilities
- New user accounts created
- Network disruption

## ***MITRE Technique***

**ID:** T1082

**Name:** System information discovery

**Description:** An adversary may attempt to get detailed information about the operating system and hardware, including version, patches, hotfixes, service packs, and architecture. Adversaries may use the information from System Information Discovery during automated discovery to shape follow-on behaviors, including whether or not the adversary fully infects the target and/or attempts specific actions.

**More info:** <https://attack.mitre.org/techniques/T1082/>

# Milestone *m4*

## Attack Step *a8*

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**Name:** Automated Collection as used by the malware

**Description:** Automated collection techniques are employed by APT28 through the utilization of the "Jaguar Tooth" malware on vulnerable Cisco routers. This technique involves the automatic gathering of information from compromised systems without direct human intervention. The exploitation process commences with the leveraging of vulnerabilities, such as CVE-2017-6742 (Cisco Bug ID: CSCve54313), to gain access to susceptible Cisco routers. Subsequently, the "Jaguar Tooth" malware is deployed within the compromised system. Pre-written lists of Cisco IOS CLI and Tcl commands are incorporated into the "Jaguar Tooth" malware, designed to extract a variety of data types. These include system information such as software versions, interfaces, and configuration details; network topology encompassing connected devices, ARP tables, and routing information; and potentially sensitive data depending on the router's configuration, which may encompass credentials or other confidential information. Automated execution and exfiltration are facilitated by the malware, which automatically executes these commands, captures the output, and transmits the collected data via TFTP (Trivial File Transfer Protocol) to a command-and-control server controlled by APT28. The consequences of this technique include reconnaissance, enabling APT28 to acquire valuable intelligence regarding the target network's infrastructure, devices, and potential vulnerabilities. Furthermore, lateral movement is facilitated through the gathered information, which can be utilized to map out the network and identify pathways for further exploitation. Data theft is another potential consequence, as sensitive data extracted from routers could be exploited for espionage or financial gain.

### Pre-Conditions:

- The target Cisco router is running a vulnerable version of Cisco IOS or Cisco IOS XE Software.
- The malware has successfully exploited the CVE-2017-6742 vulnerability.
- The SNMP service is enabled on the target router.
- An active network connection between the malware and the target router.

### Post-Conditions:

- Exfiltrated data files
- DNS queries to malicious domains
- System instability and performance degradation
- Modified system configuration files
- Suspicious process activity logs
- Potential for further malware deployment
- Altered system event logs
- Unusual network traffic logs
- Backdoor program remnants in memory or filesystem
- Firewall rule modifications
- Data exfiltration
- Compromised system security
- New user accounts created
- Remote access to sensitive systems

## **MITRE Technique**

**ID:** T1119

**Name:** Automated collection

**Description:** Once established within a system or network, an adversary may use automated techniques for collecting internal data. Methods for performing this technique could include use of a Command and Scripting Interpreter to search for and copy information fitting set criteria such as file type, location, or name at specific time intervals.

**More info:** <https://attack.mitre.org/techniques/T1119/>

## **Attack Step a9**

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**Name:** Data from Configuration Repository: Network Device Configuration Dump as used by the malware

**Description:** Cisco IOS router configurations may be extracted by malware exploiting vulnerabilities such as CVE-2017-6742. Upon successful exploitation, malicious code (shellcode) is injected into the router's memory. This shellcode executes the "show running-config" command, retrieving the current configuration settings. The captured configuration data is subsequently transferred to a remote server via the Trivial File Transfer Protocol (TFTP). Mitigation strategies include applying Cisco IOS security updates, implementing strict access controls with strong passwords and multi-factor authentication, configuring comprehensive logging for activity monitoring, and employing network segmentation to limit potential impact.

## **Pre-Conditions:**

- A system with access to the target network device.
- Network connectivity between the compromised system and the target network device.
- The affected MIBs or OIDs are not explicitly excluded on the target network device.
- SNMP is enabled on the target network device.
- The target network device has a configuration repository.
- Pre-Conditions:
- The "Jaguar Tooth" malware must be present and operational.
- Requirements:
- The target network device is running a vulnerable version of Cisco IOS Software or Cisco IOS XE Software.

## **Post-Conditions:**

- System instability
- Compromised system integrity
- Exfiltrated data files
- Backdoor executables left on the system
- Network traffic logs showing communication with C2 server
- Event logs recording suspicious activity
- Firewall logs indicating unauthorized access attempts
- Loss of availability
- Altered configuration files
- Modified system timestamps
- Data exfiltration
- Modified system image

- New user accounts created
- Loss of confidentiality

### ***MITRE Technique***

**ID:** T1602.002

**Name:** Data from configuration repository: network device configuration dump

**Description:** Adversaries may access network configuration files to collect sensitive data about the device and the network. The network configuration is a file containing parameters that determine the operation of the device. The device typically stores an in-memory copy of the configuration while operating, and a separate configuration on non-volatile storage to load after device reset. Adversaries can inspect the configuration files to reveal information about the target network and its layout, the network device and its software, or identifying legitimate accounts and credentials for later use.

**More info:** <https://attack.mitre.org/techniques/T1602/002>

# Milestone *m5*

## Attack Step *a10*

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**Name:** Exfiltration Over Alternative Protocol: Exfiltration Over Unencrypted Non-C2 Protocol as used by the malware

**Description:** The "Exfiltration Over Unencrypted Non-C2 Protocol (TFTP)" action within the Jaguar Tooth malware framework involves the collection of sensitive device information from compromised Cisco routers via the Trivial File Transfer Protocol (TFTP). This protocol, characterized by its simplicity and lack of encryption, facilitates the transfer of data between the compromised router and a designated TFTP server, likely controlled by the attackers. Data is extracted from the compromised router through the execution of abbreviated Cisco IOS CLI commands, targeting information such as running configurations, interface details, and system versions. Subsequently, the malware configures itself to utilize a specific TFTP server IP address and URL path for exfiltration purposes. The `r` (redirect) command within executed CLI commands is employed to channel the collected device information directly to the configured TFTP server. Due to the inherent absence of encryption in TFTP, all data transmitted between the compromised router and the TFTP server is transferred in plain text format, rendering it susceptible to interception by network observers. This unencrypted transmission presents significant security implications, including the potential exposure of sensitive device information such as configurations, passwords, and system details to unauthorized parties. Furthermore, the stolen information can be leveraged for subsequent attacks, facilitating lateral movement within a network or identifying vulnerabilities exploitable by attackers. While not a traditional Command and Control (C2) channel, the TFTP server could potentially be utilized by attackers to issue commands to the compromised router, thereby expanding their control over its functionality.

### Pre-Conditions:

- An unencrypted non-C2 protocol is available for use by the malware.
- The malware has successfully exploited the vulnerability to gain access to the target system.
- A network connection exists between the compromised router and the attacker's infrastructure.
- The target system is running a vulnerable Cisco IOS version.
- Pre-Conditions:
- A compromised Cisco router with a vulnerable version of Cisco IOS.
- The "Jaguar Tooth" malware must be present on the compromised router.
- The attacker has configured the malware to exfiltrate data over an alternative protocol.
- Requirements:

### Post-Conditions:

- Presence of malicious code in router memory
- Potential for further malware deployment and lateral movement within the network
- Data exfiltration from the compromised router
- Damage to the organization's reputation
- TFTP server activity transferring stolen data
- Modified router configuration files
- Network connections to command-and-control servers
- Compromised Cisco router functionality
- Log entries indicating failed login attempts or unauthorized access
- Unusual network traffic patterns, including data transfers to unknown IP addresses

- New user accounts created on the router
- Altered system timestamps and event logs
- Loss of sensitive information stored on the router
- Increased risk of denial-of-service attacks

## **MITRE Technique**

**ID:** T1048.003

**Name:** Exfiltration over alternative protocol: exfiltration over unencrypted non-c2 protocol

**Description:** Adversaries may steal data by exfiltrating it over an un-encrypted network protocol other than that of the existing command and control channel. The data may also be sent to an alternate network location from the main command and control server.

**More info:** <https://attack.mitre.org/techniques/T1048/003>

## **Attack Step a11**

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**Name:** Automated Exfiltration as used by the malware

**Description:** Jaguar Tooth malware, employed by APT28 (also known as Fancy Bear), exploits vulnerabilities within Cisco routers to facilitate unauthorized network access and data exfiltration. Exploitation of default or weak SNMP community strings, coupled with a buffer overflow vulnerability in Cisco IOS software, enables initial access and subsequent payload deployment. The deployed payloads aim to establish persistence and escalate privileges within the compromised router environment. Sensitive network information, including ARP tables, connected device details, system configuration parameters, and software versions, is gathered through the utilization of hard-coded Cisco IOS CLI and Tcl commands. This data is subsequently exfiltrated via the Trivial File Transfer Protocol (TFTP). A backdoor mechanism is established on the compromised router, granting APT28 persistent access and control even after system reboots. Mitigation strategies encompass immediate patching of Cisco IOS and IOS XE software to address identified vulnerabilities, stringent restriction of SNMP access through the implementation of strong, unique community strings and role-based access control (RBAC), and the disabling of legacy protocols such as Telnet and SNMP v1/v2c in favor of modern encrypted protocols like SSH and SNMP v3 with robust authentication mechanisms.

## **Pre-Conditions:**

- The malware is capable of establishing this connection.
- A vulnerable Cisco router is present and accessible to the attacker.
- The attacker has successfully exploited the vulnerability (CVE-2017-6742) in the router.
- A network connection exists between the compromised router and a command and control server controlled by the attacker.
- The malware has been deployed on the compromised router.

## **Post-Conditions:**

- Log entries indicating unauthorized access attempts and successful logins
- Potential for further malware deployment and lateral movement within the network
- Data exfiltration from compromised routers
- Presence of malicious code (Jaguar Tooth) in router memory
- Modified router configuration files
- New user accounts created on the router with elevated privileges
- Unusual network traffic patterns, including connections to known malicious IP addresses

- Compromised Cisco router functionality
- Altered syslog messages reflecting suspicious activity
- Increased risk of data breaches and system outages
- Loss of network integrity and security

### ***MITRE Technique***

**ID:** T1020

**Name:** Automated exfiltration

**Description:** Adversaries may exfiltrate data, such as sensitive documents, through the use of automated processing after being gathered during Collection.

**More info:** <https://attack.mitre.org/techniques/T1020/>